

RAZVAN MIHAI POPESCU

r.m.popescu@tudelft.nl | linkedin.com/in/razvan
scholar.google.com/razvan | orcid.org/razvan | Delft, Netherlands

EDUCATION

Delft University of Technology

Ph.D. in Computer Science, AI-enabled SE

Supervisors: Maliheh Izadi, Arie van Deursen

Delft, NL

Mar. 2025 - Present

Delft University of Technology

Master of Computer Science, Artificial Intelligence

Delft, NL

Sept. 2023 - Feb. 2025

Courses: Supercomputing for Big Data, Software Architecture, Information Retrieval, Artificial Intelligence Techniques, Conversational Agents, Analytics and Machine Learning for Software Engineering, Testing and Validation for AI-Intensive Systems, Artificial Intelligence for Software Testing and Reverse Engineering, Natural Language Processing, Release Engineering for Machine Learning Applications

Delft University of Technology

Bachelor of Computer Science and Engineering

Delft, NL

Sept. 2020 - Jul. 2023

Courses: Algorithms and Data Structures, Object-oriented Programming, Software Engineering Methods, Software Quality and Testing, Big Data Processing, Data Mining, Machine Learning, Computational Intelligence, Collaborative Artificial Intelligence, Human-Computer Interaction

TECHNICAL SKILLS & LANGUAGES

Programming Languages: Python, Java, C++, JavaScript, Scala, SQL

Technologies & Frameworks: Git, Docker, AWS, Hugging Face, PyTorch, Scikit-learn, Apache Spark, Apache Kafka

Languages Spoken: Romanian (Native), English (C2), German (A2)

PUBLICATIONS

Evaluating Large Language Models in Code-Related Tasks: A Systematic Literature Review

Under Review

Razvan Mihai Popescu, Maliheh Izadi

Reliable and Developer-Aligned Evaluation of Agents for Software Engineering

FSE '26 DS

Razvan Mihai Popescu

Investigating Autonomous Agent Contributions in the Wild: Activity Patterns and Code Change over Time

MSR '26

Razvan Mihai Popescu, David Gros, Andrei Botocan, Rahul Pandita, Prem Devanbu, Maliheh Izadi

Silent Reliance or Diligent Review? Human-Agent Engagement and Interaction Patterns in GitHub

ICSE-JAW '26

Ziyou Li, Razvan Mihai Popescu, Maliheh Izadi

Automated Attention Pattern Discovery at Scale in Large Language Models (extension)

TMLR '26

Jonathan Katzy, Razvan Mihai Popescu, Erik Mekkes, Arie van Deursen, Maliheh Izadi

Automated Attention Pattern Discovery at Scale

NeurIPS '25

Jonathan Katzy, Razvan Mihai Popescu, Erik Mekkes, Arie van Deursen, Maliheh Izadi

The Heap: A Contamination-Free Multilingual Code Dataset for Evaluating Large Language Models

FORGE '25

Jonathan Katzy, Razvan Mihai Popescu, Arie van Deursen, Maliheh Izadi

An Exploratory Investigation into Code License Infringements in LLM Training Datasets

FORGE '24

Jonathan Katzy, Razvan Mihai Popescu, Arie van Deursen, Maliheh Izadi

Language models for code completion: A practical evaluation

ICSE '24

Maliheh Izadi, Jonathan Katzy, Tim van Dam, Marc Otten, Razvan Mihai Popescu, Arie van Deursen

A Study on the Impact of Common Code Structures on CodeParrot's Autocompletion Performance

TU Delft

Razvan Mihai Popescu

Doctoral Researcher

Mar. 2025 – Present

AISE-TU Delft

Delft, NL

- Presenting our latest work on agentic activity at MSR '26 (Rio de Janeiro, Brazil) and my doctoral research plan at FSE '26 (Montreal, Canada)
- Supervising 2 MSc theses on agentic refactoring and agentic issue resolution in dynamic software environments, collaborating with Meta
- Attended the [Dagstuhl Seminar on Evaluation of AI Models in Software Engineering](#) '26, Schloss Dagstuhl, Germany
- Attended the international [Machine Learning Summer School](#) (MLSS '26) @ Columbia University, New York, USA
- Attended the international [Machine Learning Summer School](#) (MLSS '26) @ Maincode, Melbourne, Australia
- Web chair for the [Membership Inference Attack Competition](#), co-located with FSE '26, Montreal, Canada
- Investigated real-world activity patterns and longitudinal code evolution of coding agents and human developers
- Analyzed human engagement with coding agents, uncovering review, integration, and collaboration patterns
- Outlined a doctoral plan proposing human-aligned, real-world evaluation of LLM-powered coding agents in SE
- Conducted a systematic literature review on evaluation techniques and experimental settings for LLMs in SE
- Reviewed for leading AI4SE venues such as ICSE, FSE, ASE, ISSTA, and TOSEM
- Supervised 4 BSc theses on the impact of data smells on LLM code generation
- Web chair for the [AgenticSE Workshop](#), co-located with ASE '25, Seoul, South Korea
- Participated in the [SEN Symposium](#) '25 and [AIWare BootCamp](#) '25, Amsterdam, The Netherlands
- Presented The Heap at FORGE, co-located with ICSE '25, Ottawa, Canada

Responsible Data Science and AI Engineering Teaching Assistant

Feb. 2025 – Apr. 2025

Delft University of Technology

Delft, NL

- Supported a 400-student course, addressing student questions on responsible DSAIT
- Guided in-class discussions, encouraging students to reflect on the ethical and social aspects of DSAIT
- Graded 10+ reports and final projects, evaluating students' ability to analyze and present responsible DSAIT solutions
- Helped organize and assess 20+ final project presentations

Research Assistant

Mar. 2023 – Mar. 2025

AISE-TU Delft

Delft, NL

- Analyzed the impact of code structures on CodeParrot's completion performance, finding that common structures reduce null attention by 30% and prediction depth by 50%
- Contributed to Code4Me, an open-source IDE extension for assessing LLMs in real-world code completion scenarios
- Investigated licensing inconsistencies in LLM public training datasets, revealing up to 22% license infringements
- Contributed to developing AP-MAE, a vision transformer masked auto-encoder for identifying and reconstructing attention patterns in LLMs
- Created The Heap, a multi-lingual copyleft corpora, deduplicated against all publicly available training datasets to ensure consistency and reproducibility in LLM evaluation

Machine Learning for Software Engineering Teaching Assistant

Nov. 2024 – Feb. 2025

Delft University of Technology

Delft, NL

- Defined research proposals for student projects, focusing on LLM-driven SE
- Mentored 4 student teams, providing guidance on project implementation and academic paper writing
- Graded student submissions, evaluating both technical implementation and research quality
- Helped with organizing and assessing 15+ final project presentations

Computer Science Master Mentor

Sep. 2024 – Jan. 2025

Delft University of Technology

Delft, NL

- Guided 15 first-year CS MSc students during the Master Your Start event, helping them transition into their program
- Conducted weekly meetings to mentor students on theme selection, study planning, research groups, and course setup
- Led discussions on various academic topics, encouraging student engagement and critical thinking
- Organized social events to facilitate student networking and community engagement

Software Architecture Head Teaching Assistant

Feb. 2023 – Apr. 2023, Sep. 2024 – Nov. 2024

Delft University of Technology

Delft, NL

- Led a 260-student course, providing guidance and addressing student questions
- Mentored 10 student teams weekly in developing and documenting software architectures for socially-relevant systems

- Graded 20+ essays and proof-of-concept implementations
- Assisted in organizing and grading 20+ project presentations
- Managed software logistics for all 65 teams, improving workflow efficiency

Full-Stack Developer

Oct. 2021 – Jul. 2023

Lunar Zebro

Delft, NL

- Designed a ground segment web application for the MINAR IX Mission in collaboration with Edinburgh University
- Improved terrestrial rovers control efficiency by 80% through real-time data and video streaming
- Handled sensor data pre-processing, WebSocket & API communication, and database integration for terrestrial rovers
- Developed a ground segment web application for the IAC rover swarm, enhanced with real-time sensor data analysis, exploration mapping, object detection, and collision alerts
- Configured an asynchronous message broker for efficient control of 4 lunar rovers
- Participated in the [International Astronautical Congress](#) '22 in Paris as the only student team

Software Engineer

Apr. 2022 – Jul. 2022

Feedback-Analytics

The Hague, NL

- Collaborated on developing a word processor with form logic support, reducing report creation time by 30%
- Integrated the editor with the company's survey platform to create customized reports based on customer feedback
- Implemented an automated PDF generation system for email delivery of feedback reports
- Designed the CI/CD pipeline to optimize deployment and integration for the word processor system

PROJECTS

CNN-Based URL Phishing Detection

Apr. 2024 - Jun. 2024

- Developed a comprehensive release engineering pipeline for a CNN-based URL phishing detection model
- Contributed to key components of the pipeline, including versioning, provisioning, and containerization
- Performed testing and monitoring of the deployed model, ensuring quality control and consistency in performance

RAG for Complex QA

Jan. 2024 - Apr. 2024

- Investigated the impact of context types on answer generation in a RAG system for complex multi-hop questions
- Implemented and evaluated a RAG pipeline using Contriever and quantized LLaMa3 on 2WikiMultiHopQA dataset
- Analyzed the effects of negative and random contexts on model performance compared to relevant retrieved contexts

Conversational Movie Agent

Nov. 2023 - Jan. 2024

- Developed a conversational agent for movie recommendations integrated with memory, using the Furhat SDK
- Developed the agent's memory model (LTM and STM) and contributed to the integration of named entity recognition, sentiment analysis, and recommendation modules.
- Conducted a controlled experiment with 30 participants, showing the memory-equipped agent outperformed the memory-free agent in user satisfaction and recommendation accuracy

Prompt Augmentation for Code Generation

Sep. 2023 - Nov. 2023

- Investigated prompt augmentation techniques with partial software artifacts in LLM code generation performance
- Achieved a 43% increase in pass@1 functional performance through test case augmentation, while function signature prompts improved BLEU scores by up to 6%
- Evaluated WizardCoder and CodeGPT on the MBPP benchmark, showing that prompt augmentation enhances code generation performance, particularly in larger models